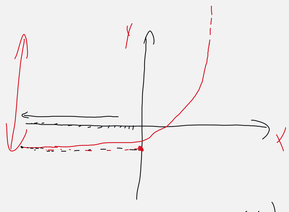
Range: (, ∞)

$\frac{4}{\sqrt{a}} = \frac{4}{0}$

$f(x) = \left(\frac{4}{\sqrt{4+x}}\right) - 2$,

$x < 4$



as $x \to -\infty$, $\frac{4}{\sqrt{4-x}} \to 0$

$f(x)$ $\frac{4}{\sqrt{4-x}} - 2$

$\frac{4-\infty}{\sqrt{4+\infty}}$

$f(3.99) = 49$

$f(3.$

$x < 4$

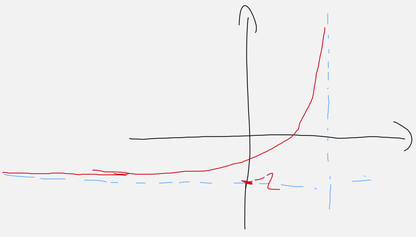
$\frac{4}{\sqrt{4-x}}$

Range: $(-2, \infty)$

	Domain	Range
$f(x)$	$x > 0$	$y < -5$
$f^{-1}(x)$	$x < -5$	$y > 0$

	Domain	Range
$f(x)$	$x < 4$	$(-2, \infty)$
$f^{-1}(x)$	$(-2, \infty)$	

$\downarrow x > -2$



(c) (i) $x = 4$

(ii) $y = -2$

$x = \frac{4}{\sqrt{4-y}} - 2$

$x + 2 = \frac{4}{\sqrt{4-y}} \quad / \cdot \sqrt{4-y}$

$(x+2)\sqrt{4-y} = 4$

$\sqrt{4-y} = \frac{4}{x+2}$

$4 - y = \frac{16}{(x+2)^2}$

$-y = \frac{16}{(x+2)^2} - 4$

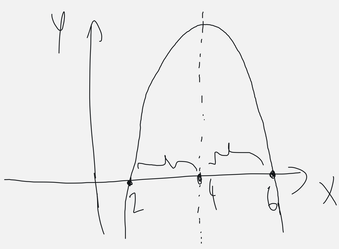
$y = -\frac{16}{(x+2)^2} + 4$

$4 - y = \frac{16}{(x+2)^2}$

$4 = \frac{16}{(x+2)^2} + y$

$4 - \frac{16}{(x+2)^2} = y$

$x_1 = -\frac{b}{2a}$



$\frac{2+6}{2} = 4$

f(20)